

USB Device Control & Monitoring Framework

What Does This Project Do?

Monitors your computer's USB ports every 5 seconds for any drive being plugged in or removed.
Checks if each USB is on your trusted whitelist — raises an alert if it is not.
Logs every event with timestamp and saves a full report you can open in Microsoft Excel.
Think of it as a security guard standing at every USB port of your computer.

**Detect USB
Plug / Remove**
Events caught
within 5 seconds

**Whitelist
Check**
Trusted vs Unknown
USB devices

**Baseline
Snapshot**
Clean starting
state saved

**CSV Report
Export**
Open in Excel
for review

6 Modules Inside usb_monitor.py

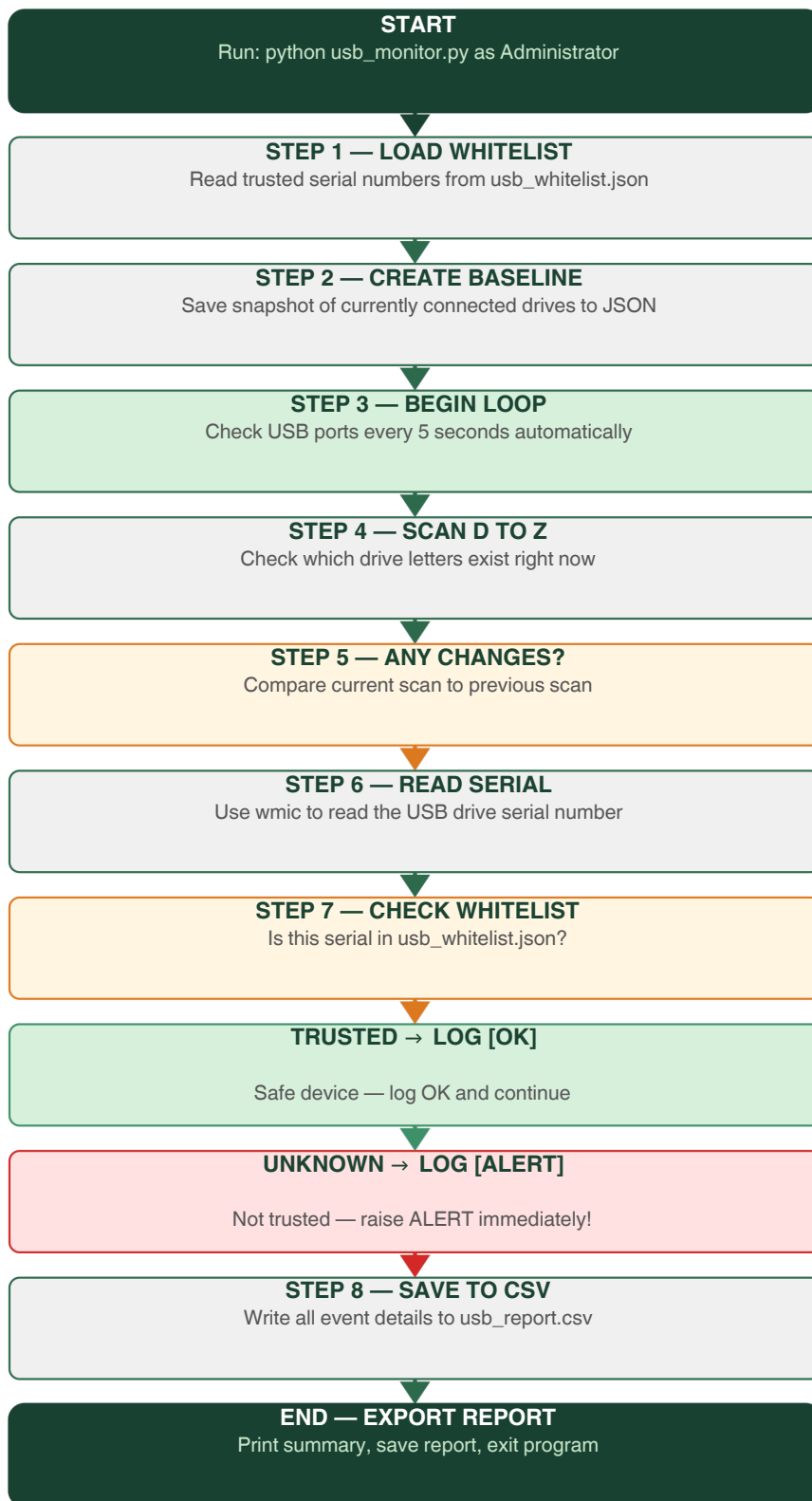
1	Configuration Scan interval,filenames, max scans — all settings in one place
2	Helper Functions get_timestamp()andlog() — shared utilities used everywhere
3	USB Detection Scans drive letters D to Z using the Windows wmic command
4	Whitelist System Loads,saves,checktrusted USB serial numbers from JSON
5	Baseline Takes and stores a clean snapshot of connected drives
6	Change Detection Compares previous scan to current — finds new or removed drives

Output Files:

usb_events.log All alerts + events with timestamps	usb_report.csv Table of all USB events	usb_baseline.json Clean starting snapshot	usb_whitelist.json Trusted USB serial numbers
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How USB Monitoring Works — Step by Step

Main Process Flowchart



USB Trust System & Alert Level Details

WHITELIST CHECK FLOW

USB Drive Plugged In



Read Drive Label & Letter



Run wmic — Get Serial No.



Check usb_whitelist.json



Serial Found in List?



YES — Log [OK] TRUSTED



NO — Log [ALERT] UNKNOWN



Save Event to CSV Report

ALERT LEVELS EXPLAINED

[OK]

Trusted USB connected
Safe — device is on whitelist

[WARN]

USB drive disconnected
Note it — was this expected?

[ALERT]

Unknown USB detected!
NOT in whitelist — investigate now!

[INFO]

Scan done — no changes
Routine scan — all clear

Signs of a Suspicious USB Device

- Serial number is NOT in the whitelist
- Drive label contains suspicious words
- USB plugged in at an unusual time
- Very large capacity appearing suddenly